

Mohamed Helal

Junior Software Developer (Flutter & Spring Boot)

✉ Mohamed.helal.02468@gmail.com | ☎ +201288898944 | 📍 Cairo, Egypt
[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

Summary

Fresh Computer Science graduate (Nov 2025) experienced in **Flutter mobile apps** and **Java Spring Boot backend development**. Skilled in building full-stack mobile applications with REST APIs, cloud deployment, and containerized solutions. Eager to contribute to a growth-oriented development team and rapidly enhance expertise in modern software development.

Education

- **Monash University Malaysia** — *Bachelor of Computer Science (November 2025)*
- **Noor British International School** — *IGCSE: A-Levels (June 2020)*

Work Experience

1. **Software Developer Intern** — *Selangor Human Resource Development Centre, Malaysia*
(Nov 2024 – Apr 2025)
 - **Cloud Infrastructure & DevOps:** Architected and deployed a containerized logistics platform (Fleetbase) using **Docker** and **AWS EC2**, configuring **OSRM/OSM** services for optimized routing and server-side stability.
 - **Real-Time Mobile Features:** Developed driver-tracking features for the **Android Navigator app (Java)**, implementing **Web Sockets** for real-time data flow and **Google Maps API** for live ETA/ECT calculations.
 - **AI & RAG Engineering:** Engineered AI-driven chatbots using **RAG** via **Dify** and **AnythingLLM**; optimized model accuracy through hyperparameter tuning and custom embedding into web environments.
 - **Third-Party Integrations:** Integrated the **Twilio API** for automated SMS/Voice alerting systems and utilized **Postman** for comprehensive API testing and endpoint validation.
2. **Software Engineer Intern** — *DYODD SDN Bhd, Malaysia*
(Nov 2023 – Feb 2024)
 - Developed and integrated **LLM-based chatbots** using **OpenAI API** and **Dify**, including dataset preparation and fine-tuning for domain-specific responses.
 - Automated business workflows using **Google Apps Script**, integrating Google Forms, Sheets, and WhatsApp APIs for real-time reporting and notifications.
 - Contributed to **CRM system development**, implementing new features (e.g., follow-up emails), fixing production bugs, and extending backend logic using APEX.
 - Deployed and tested features in **Salesforce** production, ensuring system stability and resolving issues post-deployment.

Technical Skills

1. **Languages:** Java, Dart, Python (NumPy, Matplotlib, PyTorch), SQL
 2. **Frameworks & Tools:** Spring Boot, Flutter, Android Studio, State Management (River pod)
 3. **Cloud & DevOps:** AWS, Google Cloud, Docker, GitHub
 4. **AI & ML Tools:** Fine-tuning LLMs (OpenAI, Groq), Lang Flow, Dify
-

Projects

1. Workout Tracker App (Flutter & Firebase)

Developed a cross-platform mobile application using Flutter, integrated with Firebase Authentication, Firestore, and Storage. Implemented workout tracking, progress monitoring, a rewards system, and state management using Provider and Riverpod.

2. AI Chatbot & LLM Fine-Tuning for Regulatory and Customer Support (Python)

Built and deployed AI-powered chatbots using Python, integrating OpenAI language models with platforms like Facebook Messenger. Fine-tuned LLMs on domain-specific datasets (including Strata Management Act regulations) to improve response accuracy and relevance. Designed chatbot systems for both regulatory compliance inquiries and customer support automation, ensuring context-aware and reliable responses.

3. Mobile Application Fleetbase Integration (Docker)

Deployed the backend system for the mobile application Fleetbase using Docker and integrated Google Maps API for real-time driver and order tracking within the Navigator Android application.

4. Todo List Web Application (Spring Boot + JSP + MySQL)

Developed a full-stack web application using Spring Boot, JSP, and MySQL. Implemented user authentication (sign up and login) with session-based access control. Built core features, including creating, updating, and deleting tasks, with date-based organization. Designed backend logic using Spring Boot controllers and dynamic user interfaces using JSP and Bootstrap.

5. Security Reporting System

Automated report submissions using JavaScript, Google Apps Script, Google Sheets, Google Forms, Zapier, and WhatsApp integration for real-time notifications.

Languages: *Arabic (Native), English (Fluent)*

Military Service: *Exempted*